

1. Perform basic Image Handling and processing operations on the image.

- **Read an image in python and Convert an Image to Grayscale**

Aim:

To perform basic image handling operations by reading an image in Python and converting it into a grayscale image using the OpenCV library.

Procedure:

- Install the required Python libraries (opencv-python and matplotlib) if they are not already installed.
- Import the required libraries.
- Read the input image using the cv2.imread() function.
- Convert the color image into a grayscale image using the cv2.cvtColor() function with the COLOR_BGR2GRAY option.
- Display both the original color image and the grayscale image.
- Verify that the grayscale image contains only intensity values ranging from black to white.

Code:

```
import cv2
import matplotlib.pyplot as plt

# Read image
image = cv2.imread("/Users/michellejoanna/Downloads/meee.jpeg")
# Convert BGR to RGB
rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Convert to Grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Display
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
```

```
plt.imshow(gray, cmap="gray")  
plt.title("Grayscale Image")  
plt.axis("off")
```

```
plt.show()
```

Sample Input:



Sample Output:



Result:

The input image was successfully read using Python and converted into a grayscale image using the OpenCV library.